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THE ULTIMATE AI RESEARCH AGENT AND LITERATURE REVIEW WORKSPACE

Complete Production Requirements Document (PRD) and Master Workflow Specification

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CORE VISION:

A smart, dynamic, and AI-powered literature review operating system engineered to liberate researchers, academics, and thesis students from manual data entry. It accelerates thesis and systematic review workflows by 10x through PRISMA 2020-compliant auditing, application domain taxonomy mapping, source traceability, 2D citation network topologies, and automated multi-page academic report generation.

PHASE 0: USER ROLES, AUTHENTICATION AND SYSTEM ADMINISTRATION (RBAC & MAINTENANCE)

0.1 ROLE-BASED ACCESS CONTROL (RBAC) ARCHITECTURE:

1. User Role (Researcher / Student / Co-Author / Reviewer):

- Workspace Operations: Create, edit, and organize research projects; build custom methodology clusters, application domains, and taxonomies.
- Data Ingestion and PDF Handling: Drag-and-drop batch upload of PDFs, DOI metadata pairing, and domain/keyword tagging.
- Analytical Workspace: Access 50/50 Split-Screen viewer or make dynamic Split-Screen viewer or Full Screen viewer, live math formula editor, and Visual Bounding Box (Math OCR and Table-to-CSV).
- AI Extraction and Dual Review: Execute Prompt-to-Column AI extraction, review and confirm AI Generated vs Human Verified states.
- Screening and Synthesis: Perform PRISMA 2020 inclusion/exclusion screening; generate 10 to 15 page AI research report drafts.
- Export and Sharing: Export Overleaf-ready LaTeX booktabs, BibTeX, Excel, and SVG graphs; generate secure public/supervisor read-only links.
- Quota Monitoring: View personal monthly AI token consumption, PDF storage usage, and active subscription status.

2. Admin Role (Platform Administrator / Lab Director / Maintenance):

- User and Organization Management: User CRUD (Create, Read, Update, Deactivate/Ban); assign user roles; manage university/institutional lab licenses; allocate student seat packages.
- AI Gateway and API Fleet Management: Configure LLM endpoints (Gemini, Claude, GPT, Local models); manage API keys for academic services (OpenAlex, CrossRef, Semantic Scholar); monitor rate limits and set token budgets per user/tier.
- Global Taxonomy and Master Templates: Manage pre-built benchmark templates (e.g., Default 6-Pillar Feature Selection template, NLP Benchmark template); curate preloaded seed paper libraries and standardized dataset/domain vocabularies.
- System Maintenance and Health Monitoring: Real-time dashboard for server load (CPU, Memory, Storage), database health (MongoDB/PostgreSQL connections), background task queues (OCR and PDF extraction workers), and 1-click Maintenance Mode Toggle.
- Audit Logs and Security Governance: Track system-wide error logs, security events, API latency metrics, automated database backups, and data retention policies compliant with academic privacy standards (GDPR/FERPA).

Recommendations for Phase 0:

1. Granular Collaboration Sub-Roles: Support fine-grained project permissions:

- Owner: Full control over project, billing, deletion, and sharing.
 - Editor (Co-Author): Can upload PDFs, extract columns, edit cells, suggest to delete papers or move them to archive.
 - Reviewer (Advisor/Supervisor): Can perform blind screening and leave comments/annotations without modifying extracted table data.
 - Viewer (Public Guest): Read-only access with interactive source-quote highlighting.
2. Admin Telemetry and Live Error Radar: An admin dashboard displaying real-time API uptime, AI extraction failure rates, and token cost tracking per institution.
 3. Automated Database Backup and Restore Pipeline: Daily scheduled snapshots with 1-click point-in-time recovery for disaster management.

PHASE 1: THE INTELLIGENT RESEARCH DASHBOARD AND COMMAND CENTER (COCKPIT UI)

1.1 DUAL-PERSPECTIVE DASHBOARD ARCHITECTURE:

A. Researcher / Student Mission Control (User View):

1. Project Health and Thesis Velocity Ring:
 - Live animated progress circle showing extraction readiness (e.g., "Thesis Benchmark 86% Ready - 43/50 papers deeply analyzed, 16 critical research gaps identified, 28 datasets mapped").
 - Quick indicators: Total Papers Cataloged, Screened vs Excluded, Extracted vs Pending.
 - Personal Resource Quota Widget: Monthly AI extraction token usage, PDF storage capacity, and subscription status.
2. Multi-Pillar Taxonomy Matrix Cards:
 - High-density interactive cards representing active methodology clusters (e.g., Pillar 00: Surveys, Pillar 01: Filters, Pillar 02: Wrappers, Pillar 03: Embedded, Pillar 04: Hybrids, Pillar 05: Specialized).
 - Displays live paper counts, pinned archetype mathematical formulas (such as Rough Set Degree of Dependency: $\gamma_B(D) = |\text{POS}_B(D)| / |U|$), and sub-cluster tags.
 - 1-click navigation into the 50/50 Split-Screen Workspace or Master Table.
3. Application Domain Distribution & Filter Radar:
 - Interactive visual breakdown of papers categorized by Application Domains (e.g., Cybersecurity: 46 papers, Bioinformatics/Genomics: 84 papers, Healthcare: 62 papers, Computer Vision: 51 papers, IoT/Sensors: 38 papers, NLP/Text: 29 papers, General Benchmark: 77 papers).
 - 1-click domain filter badges to instantly focus the entire dashboard on a specific research domain.
4. Living Review Feed and Preprint Radar (Elicit / ResearchRabbit Engine):
 - Live AI-powered scholarly feed scanning arXiv, IEEE Xplore, PubMed, and CrossRef.
 - Real-time notification cards: "3 New Preprints found matching mRMR and Swarm Optimization in Cybersecurity. Click to Ingest."
5. Interactive Keyword Radar and Domain Word Cloud:
 - Dynamic high-frequency keywords: [information-gain], [rough-set], [mutual-information], [high-dimensional], [bioinformatics], [network-intrusion], [microarray-classification].
 - Clicking any keyword instantly filters all project papers and methodology clusters.
6. Recent Activity and Audit Trail Timeline:
 - Chronological activity feed: "Mostafa Kamal verified 4 formula extractions in Pillar 01", "AI Agent auto-filled Time Complexity on 12

papers", "Supervisor Dr. Rahman approved PRISMA screening".

7. Quick-Action Launchpad:

- Primary action floating dock: [+ New Project], [Batch Upload PDFs], [Run Semantic Search], [Open 2D Litmap Graph], [Generate PRISMA Flowchart], [Export Overleaf LaTeX].

B. System Operations and Telemetry Command Center (Admin View):

1. Platform-Wide Operational Metrics:

- Total Registered Users, Active University Lab Licenses, Total Stored PDFs, Total Daily Extractions.

2. AI Gateway and Cost Telemetry Monitor:

- Real-time token consumption metrics across Gemini, Claude, and GPT models.
- Live API latency gauges, rate-limiting headroom, and token spend per department/institution.

3. System Health and Background Queue Radar:

- Server CPU, Memory, Disk I/O load, and database connection pool health.
- Background OCR and extraction worker queue throughput (e.g., "8 PDFs currently processing in background").
- 1-Click Maintenance Mode Toggle and Emergency Traffic Throttler.

Recommendations for Phase 1:

1. Customizable Widget Grid (Modular Dashboard): Drag-and-drop widget layout allowing researchers to rearrange cards, timeline, domain radar, keyword cloud, and velocity rings based on personal preference.
2. Dark / Light Glassmorphic Theme Switcher: Sleek, high-contrast dark mode tailored for late-night research, with instant theme memory persistence.
3. Multi-Project Quick Switcher: Seamless global project dropdown at the top header to toggle between multiple research projects in milliseconds.

PHASE 2: PROJECT SETUP AND DYNAMIC TAXONOMY & APPLICATION DOMAIN MANAGEMENT

2. Dynamic Clustering, Taxonomy and Application Domain Management:

- Users can create custom clusters/categories tailored to their thesis topology.
- Complete flexibility: Clusters and Domains can be Added, Renamed, or Deleted at any time.

Recommendations for Phase 2:

1. Color and Icon Badge Customizer:

- Allow users to assign distinct hex badge colors (e.g., Sky Blue for Filters, Purple for Wrappers, Emerald for Embedded, Amber for Hybrids, Rose for Specialized) and academic domain icons.
- Allow users to create and customize their own clusters and taxonomies.

2. Standard Application Domain Registry:

- Provide a built-in pre-configured registry of academic Application Domains with distinct domain icons and color tags:
 - Cybersecurity and Network Intrusion Detection (Shield icon, Red badge)
 - Bioinformatics, Genomics and Microarray Data (DNA icon, Emerald badge)
 - Healthcare, Biomedical and Clinical Diagnostics (Hospital icon, Blue badge)
 - Computer Vision, Biometrics and Image Processing (Eye icon, Purple badge)
 - Natural Language Processing and Text Mining (Document icon, Amber badge)
 - Internet of Things (IoT), Edge Computing and Sensor Networks (Globe icon, Cyan badge)
 - Financial Risk, Fraud Detection and FinTech (Chart icon, Green badge)
 - General Machine Learning and Theoretical Benchmarks (Microscope icon, Slate badge)
- Allow researchers to add custom application domains specific to their

- thesis (e.g., Agricultural Remote Sensing, Autonomous Driving, Power Systems).
3. Archetype Mathematical Formulation Attachment: Enable pinning a foundational formula (e.g., Rough Set Dependency Degree $\gamma_B(D) = |\text{POS}_B(D)| / |U|$) and variable dictionary to the cluster header so all papers in that category automatically inherit the theoretical baseline.
 4. 2-Level Nested Sub-Taxonomy Trees: Support sub-categories within major clusters (e.g., Wrapper Methods $\rightarrow$ Swarm Intelligence (PSO/GWO) vs. Evolutionary Algorithms (GA/DE)).
 5. Safe Cluster Archiving and Paper Re-routing: When deleting a cluster, prompt a modal offering to reassign papers to another cluster or move them to an Unassigned Archive without deleting underlying PDF data.

PHASE 3: PRISMA-COMPLIANT DATA COLLECTION AND AI SCREENING

3. BYOData (Bring Your Own Data), DOI Auto-Fetching and Domain Tagging:
 - Bulk Upload: Users can drag-and-drop or browse their directories to upload multiple PDF files simultaneously.
 - DOI Integration: If a user inputs a DOI or URL, the system fetches the metadata (Title, Abstract, Year, Citation Count) via Crossref/Semantic Scholar APIs.
 - Note: The system does not download the PDF automatically. The user uploads the PDF file, and the system pairs it with the fetched DOI metadata.
 - Add a simple Agentic AI API for:
 - Filling metadata of the columns.
 - Setting cluster assignments.
 - Detecting and tagging Primary and Secondary Application Domains.
 - Adding domain keywords and dataset names.

Recommendations:

1. Intelligent Deduplication and Auto-Merge Engine (Rayyan-style): Run automated fuzzy string matching (Levenshtein distance on normalized titles + exact DOI collision detection) during bulk upload. Display a Duplicate Reconciliation Modal allowing 1-click merging of duplicate records from multiple repository imports.
2. Agentic AI Auto-Classifer and Multi-Domain Tagging Pipeline: When a PDF is dropped:
 - An autonomous micro-agent computes cosine similarity between paper abstracts and active clusters, recommending the best methodology cluster with a confidence score (e.g., "96% Confidence Match $\rightarrow$ Pillar 01: Filters").
 - Automatically detects and tags the paper's Primary Application Domain (e.g., "Bioinformatics / Microarray") and any Secondary Application Domains (e.g., "Clinical Oncology Diagnostics").
 - Auto-generates 5 normalized domain keywords (e.g., [information-gain], [mrmr], [high-dimensional], [colon-tumor-microarray]).
 - Resolves publication venues (IEEE TPAMI, Elsevier KBS, Springer, Nature, ACM).
3. Local PDF Text and Metadata Extractor: Extract raw text, author affiliations, publication year, and structured abstract sections locally via pdfjs-dist to enable offline-first productivity.
4. AI PRISMA Screening (Inclusion/Exclusion):
 - Natural Language Criteria: Before papers enter the Unassigned queue, users define filtering rules (e.g., "Exclude papers published before 2020, exclude survey papers, must evaluate on Cybersecurity or Healthcare datasets").
 - Auditable Traceability: The AI scans hundreds of papers in seconds, routing non-compliant papers to an Excluded List. To strictly follow PRISMA 2020

- guidelines, it displays an exact quote or a generated reason next to the excluded paper explaining why it was dropped.
- Cluster Assignment: Approved papers can be moved to specific clusters via:
 - Drag & Drop.
 - Dropdown selection.
 - A "Select to Transfer" batch menu.

Recommendations:

1. 5-Star AI Priority Relevance Ranking (Rayyan-style): After the user manually screens the first 5 to 10 papers, an active learning classifier predicts inclusion probability for remaining papers, tagging them with 1 to 5 star relevance badges so high-impact papers float to the top of the screening queue.
2. Interactive PRISMA 2020 Flowchart Generator: Dynamically render a publication-standard PRISMA 2020 flow diagram (Identification -> Screening -> Eligibility -> Included) with real-time numeric counters, exportable in 1-click as a vector SVG, PDF, or Overleaf LaTeX TikZ figure.
3. Color-Coded Inclusion/Exclusion Keyword Radar: Auto-highlight user-defined positive keywords (e.g., benchmark, complexity, accuracy, genomics) in soft green and disqualifying terms (e.g., review only, qualitative, simulation only) in soft red within abstract screening cards.
4. Student-Advisor Double-Blind Mode: Support a blind screening toggle where co-authors/supervisors evaluate studies independently, revealing discrepancies and computing Inter-Rater Reliability (Cohen's Kappa) during final conflict resolution.

PHASE 4: THE SMART SPLIT-SCREEN WORKSPACE (DEEP EXTRACTION)

5. Split-Screen Reading and Visual Extraction:
 - Clicking a paper inside a cluster opens the Workspace.
 - Left Panel: PDF Viewer (supports zoom, pan, and highlight).
 - Right Panel: Dynamic Data Entry Panel (displays all columns, methodology cluster, application domain badges, keyword tagging, and edit options).
 - Quick Actions: Change cluster assignment, modify application domains, or update keywords directly from this view.
 - Visual Bounding Box: Users can use their mouse to draw a box (crop) over:
 - diagrams,
 - experimental tables,
 - math equations in the PDF.
 The selection is instantly saved as an image or extracted text in the corresponding cell on the right panel.

Recommendations:

1. Application Domain Metadata Widget: Dedicated domain selector in the right panel displaying color-coded domain badges (e.g., [Cybersecurity], [Bioinformatics]) with 1-click multi-tag editing.
2. Math Equation Snipping and Math OCR (SciSpace-style): Drawing a bounding box over any mathematical formulation converts pixel data into valid LaTeX code rendered live via KaTeX, while automatically generating an editable symbol dictionary table (symbol -> definition).
3. Benchmark Table-to-CSV Matrix Extractor: Cropping experimental result tables (e.g., Accuracy %, F1-Score, Execution Time) parses table cells into structured JSON/CSV data ready for direct injection into the benchmark comparison matrix.
4. Contextual Text-Selection Tooltip (Quick-Send Actions): Highlighting any paragraph in the PDF opens a non-intrusive floating menu: [Send to Core

Intuition], [Add to Key Strengths (Pros)], [Add to Critical Gaps], [Send to Math Formula], [Set Application Domain].

5. In-Viewer AI Thesis Copilot (SciSpace-style): An inline explain button ("Explain for Thesis") that translates complex technical jargon into concise, academic language.
6. Dynamic Column Management and Split Logic:
 - Add & Split: Users can create custom columns (which automatically sync across all papers in that cluster). Crucially, they can Split a column into sub-columns:
 - Example: splitting Complexity into:
 - Time Complexity (TC)
 - Space Complexity (SC)

Recommendations:

1. Type-Aware Column Schemas: Support diverse column data types:
 - Text / Rich Text (Intuition, Limitations)
 - KaTeX Math Formula (LaTeX math blocks)
 - Multi-Tag Chips (Datasets, Classifiers, Application Domains)
 - Numeric Metric (Accuracy %, Execution Time in seconds, Feature Reduction %)
 - Bullet Lists (Pros, Gaps, Open Opportunities)
2. Cluster-Wide Schema Sync with Retroactive Migration: Splitting or adding a column in one paper instantly propagates the schema to all papers in that cluster, with the AI retroactively splitting existing composite text into the newly formed sub-columns.
3. Column Reordering and Visibility Toggles: Allow drag-and-drop column reordering and checkbox-based visibility toggling to customize workspace density.
7. AI Auto-Fill and Human-in-the-Loop Verification:
 - The Magic: As soon as a user adds a new column, the Research Agent reads the entire PDF and auto-fills the extracted values.
 - Verification Badge: AI-populated data receives a yellow "AI Generated" badge. Once the user reviews the data and clicks the checkmark or edits the text, it converts to a green "Human Verified" badge.

Recommendations:

1. Confidence Scores and Low-Confidence Flags: Pair the AI Generated badge with an extraction confidence indicator (e.g., 96% High Confidence vs. 58% Low - Needs Review). Low-confidence items trigger a direct verification prompt.
2. Custom Natural Language Prompt Builder (Elicit-style): Allow users to customize the exact extraction prompt per column (e.g., "Extract the exact objective function minimized, including penalty terms" or "List all benchmark datasets with sample sizes and domain categories").
3. Human-in-the-Loop Dual Review Workflow: Provide a "Verify All" batch auditor where researchers can cycle through AI-extracted snippets side-by-side with original PDF paragraphs in seconds.

PHASE 5: MASTER MATRIX AND HIGH-SCALE ANALYSIS

8. Cluster-Based Dynamic Spreadsheet:
 - Navigating back to the cluster's main page reveals a massive, dynamic Excel-like table capable of loading thousands of data points smoothly.
 - Inline Edit: Users can click on any cell in the table to quickly update or edit values without needing to open the split-screen view.

Recommendations:

1. Multi-Mode Presentation View Switcher: Provide 4 switchable perspectives:
 - Cards View: Deep analytical cards with collapsible KaTeX variable drawers and application domain badges.
 - Benchmark Table View: High-density virtualized spreadsheet with multi-column sorting and pinning.
 - Heatmap Matrix View: 2D performance grid (Algorithms vs. Datasets vs. Domains vs. Accuracy).
 - 2D Litmaps Graph View: Interactive citation timeline landscape.
2. Global Application Domain Filter Pills: A top filter bar displaying domain pills (e.g., [All Domains (387)], [Bioinformatics (84)], [Cybersecurity (46)], [Healthcare (62)], [Computer Vision (51)], [IoT / Edge (38)]). Clicking any pill filters the entire matrix in real-time.
3. Cross-Domain Generalizability Analysis: Compare algorithm robustness across different application domains (e.g., evaluating how mRMR or GA wrappers perform on 20,000-feature sparse genomic data versus 41-feature network packet data).
4. Universal Multi-Field Search and Filter Pills: Real-time search across titles, authors, algorithms, mathematical symbols, dataset chips, application domains, and gap keywords.
5. Column Aggregation and Statistics: Automatic footer calculations showing total datasets evaluated, domain breakdown, average accuracy, and metric distributions across the cluster.
9. Sentence-Level Traceability (Source Citation):
 - Source Highlighting: Clicking any value inside the Master Table or the right-side panel triggers the left PDF viewer to automatically scroll to the exact source. The specific sentence the AI extracted the data from will be highlighted in yellow.

Recommendations:

1. Pixel-Accurate PDF Anchor Coordinates: Store page numbers and normalized bounding box coordinates [pageNumber, x, y, width, height] alongside extracted data for sub-second jump-to-page navigation and precision yellow highlighting.
2. Audit Verification Stamp and Quote Drawer: Clicking an extraction badge opens a provenance drawer displaying: "Extracted from Page 6, Paragraph 3, IEEE TPAMI 2024. Verified by Researcher on Aug 2026."
3. Zero-Hallucination Academic Trust: If an extracted claim lacks an exact verifiable sentence anchor in the PDF, flag it with an Unanchored Warning requiring researcher confirmation.

PHASE 6: INSIGHTS, REPORT GENERATION AND ACADEMIC EXPORT

10. Automated Unique Value Summary (Cluster Insights):
 - At the bottom of the table (or in a sidebar), the system analyzes the column data and auto-renders a bulleted summary of unique values.
 - Example: "Datasets Used: NSL-KDD (Cybersecurity), Colon Microarray (Bioinformatics)" or "Key Limitations: High memory usage in genomic data, Slow convergence on continuous features".

Recommendations:

1. Frequency Distribution Badges: Display frequency count tags next to every unique element (e.g., NSL-KDD (28 papers), MNIST (21 papers), CIC-IDS2017 (12 papers), Colon Microarray (18 papers)).
2. Methodology Pillar vs Application Domain Cross-Tabulation Heatmap:
 - Generates a 2D matrix mapping Methodology Pillars (Filters, Wrappers,

- Embedded, Hybrids, Rough Sets) against Application Domains (Cybersecurity, Genomics, Healthcare, Vision, IoT, NLP).
- Highlights under-explored algorithmic intersections (e.g., "Rough Sets + Deep GNNs have 42 studies in Healthcare, but 0 papers in IoT Cybersecurity - Prime Thesis Opportunity!").
- 3. Consolidated Methodological Trade-Off Matrix: Automatically cross-tabulate Pros vs. Gaps across all papers in the cluster to identify structural bottlenecks across domains.
- 11. AI Research Reports (Automated Thesis Drafts):
 - Generate Report: Clicking this button prompts the AI to synthesize all papers in the cluster, the custom column splits, domain distributions, and the unique summaries to write a structured, 10 to 15 page Literature Review Draft (in paragraph format).
 - Every claim in the report includes an inline citation. Clicking the citation jumps directly to the specific line in the original PDF.

Recommendations:

1. Modular Thesis Chapter 2 Outlines: Offer pre-formatted academic chapter structures with domain breakdowns:
 - Section 2.1: Mathematical Archetypes and Theoretical Formulations
 - Section 2.2: Comprehensive Benchmark Taxonomy (Filter vs Wrapper vs Hybrid)
 - Section 2.3: Cross-Domain Empirical Performance & Complexity Comparison (Bioinformatics vs Cybersecurity vs Healthcare)
 - Section 2.4: Identified Domain Limitations and Novel Thesis Contribution Roadmap
2. Cross-Paper Multi-Document Synthesis Q&A (Elicit/SciSpace Engine): An interactive research chat allowing multi-paper comparative queries (e.g., "Compare the computational time complexity bottlenecks of Swarm Intelligence vs. Rough Set methods on high-dimensional genomic microarray data").
3. Clickable Interactive In-Text Citations: Every paragraph claim includes linked citations ([1], [2], Peng et al., 2005) that jump directly to the verified source quote in the split-screen PDF.
12. Custom Export and Transparent Sharing:
 - Selective Export: Users can export data by cluster, by application domain, or select specific columns to download as Excel (.xlsx), CSV, RIS, or DOCX files.
 - Public Link: Users can generate a Read-only or Editable link to share the entire split-screen dashboard with supervisors or co-authors - no account creation required for the guest.

Recommendations:

1. Overleaf-Ready LaTeX Booktabs and BibTeX Exporter: Export publication-ready LaTeX table snippets (`\begin{table}`, `\toprule`, `\midrule`, `\bottomrule`) and a clean, validated .bib citation library ready for instant Overleaf integration.
2. High-Resolution Vector SVG / 300 DPI PNG Graph Export: 1-click download of the 2D citation network map and PRISMA 2020 flowchart formatted with legends for direct inclusion in thesis figures.
3. Full Portable JSON Workspace Backup: 1-click export/import of the entire project state, column schemas, domain taxonomies, extracted formulas, and annotations for backup or migration.

ADDITIONAL MACRO-TOPOLOGY ENGINE (LITMAPS + RESEARCHRABBIT INTEGRATION)

13. Interactive 2D Citation Network and Algorithmic Lineage Trees:

- 2D Citation Landscape Canvas: Interactive timeline visualization where X-axis = Publication Year (1991 to 2026), Y-axis = Citation Impact, and nodes are color-coded by Methodology Pillar or Application Domain.
- Seed-Paper Snowballing Engine: Designate a foundational seed paper (e.g., Peng et al. 2005 - mRMR) to automatically discover top backward references and forward citing papers via OpenAlex / Semantic Scholar.
- Tri-Directional Literature Discovery Hops:
 - Earlier Work: Foundational ancestral algorithms.
 - Similar Work: Peer benchmark approaches within the same application domain.
 - Later Work: Recent 2024-2026 algorithmic extensions and hybridizations.
- Co-Authorship and Research Lab Collaboration Graph: Visual network mapping top publishing authors and research groups in the field.

PHASE 7: THE PUBLIC-FACING HOME PAGE AND MASTER SHOWCASE PORTAL

7.1 Public Landing Page and Master Corpus Architecture:

1. Hero Showcase and Value Proposition:

- Headline: "The Ultimate AI Research Agent and Literature Review Workspace".
- Core Tagline: "Accelerate your academic thesis and systematic reviews by 10x with PRISMA 2020 screening, application domain classification, sentence-level PDF quote traceability, live math formula extraction, and 2D citation network topologies."
- Interactive Hero Playground: A live mini 50/50 split-screen demo allowing visitors to crop a sample math formula (e.g., Peng et al. mRMR) and see it convert to clean LaTeX code and highlight its source sentence in yellow without logging in!
- Primary CTAs: [Explore Master Benchmark Corpus (387+ Papers)], [Start Free Workspace], [Watch 2-Min Demo].

2. Master Benchmark Corpus, 6-Pillar and Application Domain Showcase:

- 6 Methodology Pillars:
 - Pillar 00: Surveys and Systematic Reviews (59 Papers)
 - Pillar 01: Filter-Based Feature Selection (65 Papers)
 - Pillar 02: Wrapper and Metaheuristic Optimization (98 Papers)
 - Pillar 03: Embedded and Sparse Learning (32 Papers)
 - Pillar 04: Hybrid and Ensemble Systems (86 Papers)
 - Pillar 05: Specialized and Emerging Paradigms (47 Papers)
- Application Domains Catalog:
 - Cybersecurity (46 Papers - NSL-KDD, CIC-IDS2017)
 - Bioinformatics and Genomics (84 Papers - Colon, Leukemia, Lymphoma, Isolet)
 - Healthcare and Biomedical (62 Papers - Arrhythmia, Breast Cancer)
 - Computer Vision and Biometrics (51 Papers - MNIST, CIFAR-10)
 - IoT and Sensor Networks (38 Papers)
 - NLP and Text Mining (29 Papers)
 - General Benchmarks (77 Papers)
- Corpus Stats Banner: 387 Chronologically Indexed Papers (1991 to 2026) | 1,200+ Extracted Formulations | 85 Benchmark Datasets | 8 Major Domains.

3. Quintuple-Platform Competitive Advantage Grid:

- Comparison matrix uniting Elicit + Litmaps + ResearchRabbit + Rayyan + SciSpace into ONE unified tool.

4. Interactive 2D Citation Network Canvas (Live Demo):

- Embedded WebGL/D3 2D citation graph on the home page for interactive exploration.

5. Academic Trust, Social Proof and Starter Pack Downloads:
 - Badges: PRISMA 2020 Compliant, Overleaf / LaTeX Ready, OpenAlex Verified.
 - Free downloads: Overleaf LaTeX Booktabs templates, BibTeX seed libraries, and PRISMA Flowchart SVGs.

Recommendations for Phase 7:

1. Interactive Sandbox Mode: Test split-screen PDF editor and math OCR directly on the landing page with sample papers from different domains.
2. Live Dynamic Counter Badges: Display total papers extracted, domains analyzed, and research hours saved.
3. Interactive FAQ Section: Clarify sentence-level traceability, zero hallucinations, data privacy, and Overleaf exports.

PHASE 8: THE COMPREHENSIVE "ABOUT THE PROJECT" MANIFESTO AND ACADEMIC OVERVIEW

8.1 Academic Identity and Master Thesis Manifesto:

1. Project Identity and Purpose:
 - Official Title: Feature Selection and Literature Review Operating System (Master Thesis Knowledge Corpus).
 - Core Research Problem: The exponential growth of academic literature (2.5M+ papers/year) has broken traditional manual literature review workflows. Researchers suffer from:
 - Information overload and unmanageable PDF folders across fragmented domains.
 - Subjective, non-reproducible paper screening prone to bias.
 - Tedious, error-prone manual extraction of complex mathematical formulas and Big-O computational complexities.
 - AI hallucinations in generic chatbots (ChatGPT/Claude) that summarize without verifiable PDF text anchors.
 - Fragmented toolchains forcing researchers to juggle 5+ disconnected platforms.
 - The Breakthrough Solution: A unified, offline-first, LaTeX-native, and PRISMA-compliant AI Research Workspace that bridges Macro Citation Topologies (Litmaps/ResearchRabbit) with Micro Semantic Extraction (Elicit/SciSpace), Multi-Domain Benchmarking, and Systematic Bias-Controlled Screening (Rayyan).
2. The Master Benchmark Corpus Architecture (387 Deep-Indexed Papers):
 - Comprehensive chronological mapping spanning 35 years of machine learning and feature selection innovation (1991 to 2026):
 - 00 Surveys and Reviews (59 Studies): Foundations, taxonomies, curse of dimensionality, and evaluation metrics.
 - 01 Filter Methods (65 Studies): Mutual Information (mRMR), ReliefF, Laplacian Score, Fisher Score, Pearson Correlation.
 - 02 Wrapper and Metaheuristics (98 Studies): Genetic Algorithms, BPSO, Grey Wolf Optimizer, Ant Colony, Whale Optimization.
 - 03 Embedded and Sparse Learning (32 Studies): Lasso L1, ElasticNet, Decision Tree Gini Impurity, Regularized Forests.
 - 04 Hybrid and Ensemble Systems (86 Studies): Two-stage filter-wrappers, Pareto multi-objective optimization, stacked ensembles.
 - 05 Specialized and Emerging Frontiers (47 Studies): Rough Set Theory lower approximations ($\gamma_B(D)$), Markov Blanket, Streaming features, Graph Neural Networks.
 - 85 Standardized Benchmark Datasets across 8 Application Domains: Cybersecurity (NSL-KDD, CIC-IDS2017), Vision (MNIST), Bioinformatics and

Genomics (Colon, Leukemia, Isolet), Healthcare (Arrhythmia).

3. Methodological Rigor and First-Class Mathematical Architecture:
 - Math Formula Typesetting: Native rendering of objective functions, fitness evaluations, regularization penalties, and Rough Set boundary regions.
 - Variable Dictionary Standard: Standardized symbol definition tables (symbol -> definition -> domain).
 - Big-O Computational Complexity Taxonomy: Detailed asymptotic analysis cataloged per technique.
4. Technology Stack and Architectural Pillars:
 - Frontend Engine: Vanilla CSS Modular Design System, PDF.js Vector Canvas, Tabulator.js Dynamic Spreadsheet, KaTeX Live Typesetting, D3.js 2D Graph.
 - Backend and Storage: IndexedDB (Client offline storage), MongoDB (Document store for nested column splits & offsets), LocalStorage.
 - Publishing Pipeline: Overleaf LaTeX booktabs, BibTeX .bib, Multi-Sheet Excel, and Vector SVG graphics exporter.
 - Author and Context: Mostafa Kamal • Final Year Master's Thesis Project • Open Science and Reproducible Research.

Recommendations for Phase 8:

1. Interactive "Meet the Benchmark" Explorer: Search and inspect any of the 387 foundational papers across all 8 application domains with live equation popups.
2. Downloadable Thesis Manifesto PDF: 1-click download of the complete theoretical specification as an academic whitepaper.
3. Contributor and Citation Guide: Pre-formatted BibTeX and APA citations for citing this Literature Review Ecosystem in future dissertations.

PHASE 9: ACCOUNT AUTHENTICATION, REGISTRATION AND INSTITUTIONAL ONBOARDING

9.1 Modern Academic Authentication and Onboarding Architecture:

1. Visual Split-Screen Auth and Showcase Layout:
 - Left Showcase Panel (Project DNA and Value Proposition):
 - High-contrast animated glassmorphic showcase.
 - Core pillars spotlight: PRISMA 2020 Compliance Engine, Sentence-Level PDF Traceability, Live Formula OCR, 2D Citation Litmap Graph, 387-Paper Preloaded Benchmark Corpus across 8 Application Domains.
 - Rotating verified research quotes and advisor testimonials.
 - Right Authentication Hub (Sign-In and Registration Form):
 - Academic Single Sign-On (SSO):
 - Continue with ORCID iD (The global persistent digital identifier for academic researchers).
 - Continue with Google / Google Scholar.
 - Continue with GitHub (For data scientists and ML developers).
 - Institutional Edu SSO (Shibboleth / InCommon) (Log in with university student/faculty credentials).
 - Standard Credentials and Passwordless Magic Links:
 - Secure email + password registration with Argon2/bcrypt hashing.
 - Passwordless "Magic Sign-In Link" sent directly to inbox for 1-click password-free access.
2. Personalized Academic Onboarding Wizard (First-Time User Flow):
 - Step 1: Researcher Persona Selection:
 - Master's Student (Thesis Mode - Pre-configures Chapter 2 Templates and

- LaTeX Booktabs).
 - PhD Candidate / Postdoc (Comprehensive Scoping and Systematic Review Mode).
 - Professor / PI (Reviewer / Supervisor Lab Mode - Read-Only Audit and Blind Review).
 - Industry R&D Practitioner (Algorithmic Benchmarking Mode).
 - Step 2: Research Discipline and Application Domain Starter Seed:
 - Choose starting domain:
 - Computer Science and Machine Learning (Pre-loads 6-Pillar Feature Selection Benchmark Corpus)
 - Bioinformatics and Computational Biology / Genomics
 - Cybersecurity and Network Intrusion Detection
 - Healthcare and Clinical Diagnostics
 - Computer Vision and Image Processing
 - Blank Custom Taxonomy and Domain Setup
 - Step 3: Reference Manager Integration:
 - 1-click OAuth connect with Zotero, Mendeley, or Overleaf Git Sync.
 - Step 4: Interactive Quick-Tour and Sandbox Project Launch:
 - Instant launch into a pre-configured sample project with 3 benchmark papers to practice PDF math cropping and quote clicking in 60 seconds.
3. Enterprise Security, Privacy and Token Quota Governance:
- Multi-Factor Authentication (MFA / 2FA): Support for TOTP Authenticator Apps (Google Authenticator, Authy) and biometric WebAuthn (TouchID, FaceID, Passkeys).
 - Encrypted JWT Sessions: Stateless JSON Web Tokens stored in HttpOnly, SameSite=Strict cookies with automatic silent refresh.
 - Academic Data Sovereignty and FERPA/GDPR Compliance: Encrypted local PDF storage, zero unauthorized training on user PDFs, and instant account data export/erasure.
- Recommendations for Phase 9:
1. Verified .edu Instant Student Upgrade: Automatically detect .edu or university domains upon sign-up to apply the 50% Academic Discount tier instantly without manual support tickets.
 2. Supervisor Invitation Loop: During onboarding, allow students to enter their Thesis Advisor's email to automatically send a "Supervisor Read-Only Collaboration Invite" with a pre-configured audit link.
 3. Offline Guest Mode with Local IndexedDB: Allow students to use the core workspace locally without creating an account, with a gentle "Create Free Cloud Account to Sync Across Devices" prompt.

BUSINESS MODEL AND PRICING STRATEGY (DRAFT)

- Basic (Free):
 - Unlimited local PDF uploads, basic column additions, manual data entry, KaTeX formula rendering, and standard CSV export.
- Plus (15 USD / month):
 - AI PRISMA Screening, Column Splitting, DOI Auto-fetch, limited AI Auto-fill credits.
- Pro (40 USD / month):
 - Sentence-level Traceability, Table/Figure Bounding Box Extraction (Math OCR), AI Research Report Generation (10+ page drafts), 2D Litmap Citation Graph, and Unlimited AI Extraction.

- Enterprise (Custom):
University/Lab integrations, unlimited team collaboration, dedicated API access, and private cloud deployment.

Recommendations for Pricing Strategy:

1. Academic Student Annual Plan (50% Off): Special verified .edu tier at 8 USD / month (billed annually) to capture global Master's and PhD student markets during thesis writing seasons.
2. Pay-As-You-Go AI Token Top-Ups: Small on-demand credit packs (e.g., 5 USD for 100 deep-extracted papers) for researchers working on massive scoping reviews without requiring enterprise contracts.
3. Departmental / Lab Seat Packs: Discounted multi-seat packages (5 seats for 150 USD / month) tailored for academic research labs and university departments.

NEXT IMPLEMENTATION STEPS

The Literature Review Ecosystem is fully specified as an enterprise-grade "Research Operating System" with complete RBAC, Command Dashboard, Deep Split-Screen Extraction, Master Matrix, Citation Topology, Public Showcase Home Page, Comprehensive Project Manifesto, and Academic Authentication/Onboarding Portal.

Recommended Technical Starting Points:

1. MongoDB / Data Structure Engine: Finalize the document schema supporting nested column splits, KaTeX math symbols, bounding box coordinates [x, y, w, h], Application Domain tags, RBAC User/Admin permissions, and dual-verification badges.
2. Frontend UI/UX and Split-Screen Workspace: Implement the 50/50 Split-Screen layout (with dynamic Full Screen / Split Screen toggling) with PDF.js, Visual Bounding Box cropping, and dynamic Tabulator.js spreadsheet rendering.

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